

SANTHOSH REDDY KATASANI VENKATA

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EDUCATION

University of Illinois Chicago

May 2026

Master of Science in Computer Science

Courses: *Neural Networks, Advanced Machine Learning, Deep Learning for Computer Vision, ML on Graphs, Visual Data Science, NLP, Pattern Recognition, Computer Algorithms, Big Data Mining*

SRM Institute of Science and Technology

May 2024

Bachelor of Technology in Computer Science

EXPERIENCE

Research Assistant, UI Health

Jun 2025 – Present

Developed an AI-powered recipe recommendation system using a Retrieval-Augmented Generation (RAG) pipeline with OpenAI LLMs to generate personalized cancer-friendly recipes based on user constraints (calories, cuisine, dietary preferences). Built and integrated the end-to-end system with a FastAPI (Uvicorn) backend, FAISS-based retrieval, and a React (Vite) frontend, designed for cloud deployment on Azure.

Software Development Intern, Soft Logic Systems

2023 June - 2023 December

Contributed to application development using Python and Java, working with REST APIs and MySQL databases in an Agile team environment. Assisted with debugging, API testing, and data handling workflows, strengthening software engineering practices for production systems.

PROJECTS

1. Distribution-Aware Visual Prompt Tuning for Lung Cancer Risk Prediction

Dec 2025

Built a slice-based Vision Transformer (ViT-B/14) pipeline for 1-year lung cancer risk prediction from low-dose CT scans, integrating MedSigLIP medical foundation models, anatomical positional encoding, and attention-based pooling. Applied Distribution-Aware Visual Prompt Tuning (DA-VPT) for parameter-efficient fine-tuning on a frozen backbone, improving AUC-ROC from **~0.65 to ~0.77**.

Technologies: *Python, PyTorch, HuggingFace, MLflow*

Architecture: *Vision Transformers, Prompt Tuning, Attention-Based Pooling*

2. Neural Machine Translation

May 2025

Implemented Seq2Seq LSTM and Transformer-based models for French-to-English translation, including tokenization, padding, vocabulary indexing, and teacher forcing. Evaluated translation quality using BLEU scores and improved output fluency through attention mechanisms and optimized decoding strategies.

Technologies: *Python, TensorFlow/Keras, NLTK*

Architecture: *Encoder-Decoder, LSTM, Transformer, Attention*

3. De-Expression Learning

Jan 2025

Developed a computer vision pipeline for facial de-expression using residue learning and a U-Net-based conditional GAN architecture. Designed custom loss functions and trained models to generate expression-neutral facial images from expressive inputs while preserving identity features and structural facial consistency.

Technologies: *Python, TensorFlow, OpenCV*

Architecture: *Conditional GANs, U-Net, Residue Learning*

4. Learning Based Airline Delay Estimation

Oct 2024

Developed a predictive model to classify flight delays using large-scale flight and weather data, performing data analysis and applying deep learning algorithms to improve decision-making. Handled data cleaning, missing values, and feature engineering, and identified key delay-causing factors through exploratory data analysis.

Technologies: *Python, TensorFlow, PySpark, Pandas, Scikit-Learn, Tableau, Matplotlib*

TECHNICAL PROFICIENCIES

Languages:	Python, Java, HTML, SQL
ML & DL:	PyTorch, TensorFlow, Keras, Transformers, Vision Transformers
ML Systems:	FastAPI, FAISS, HuggingFace, MLflow
Data & Cloud:	Pandas, NumPy, PySpark, Azure
Dev Tools:	Git, Linux